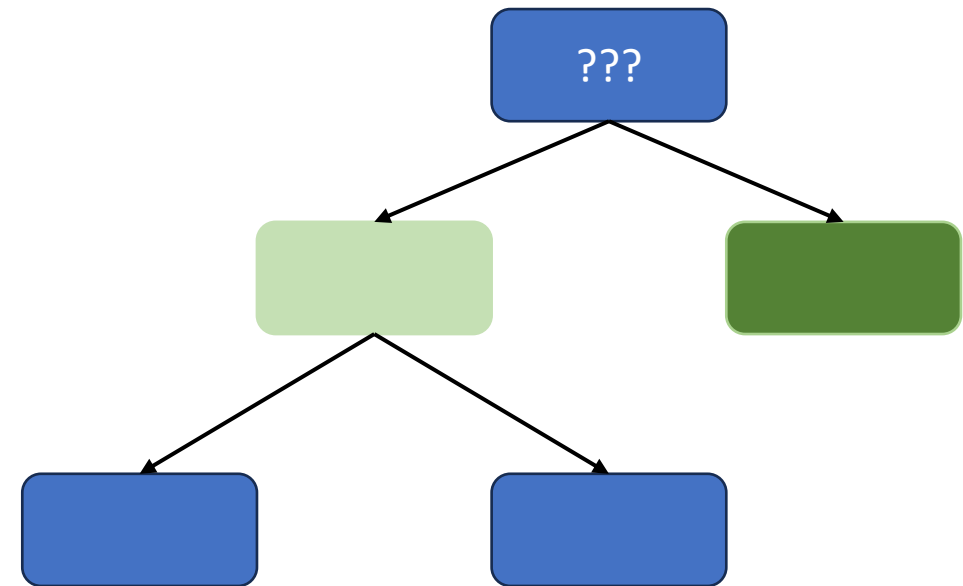


Building a decision tree

Building Tree with Gini Impurity

The first thing we do is decide whether Likes Math, Does Homework or Study Hours should be the question we ask at the very top of the tree

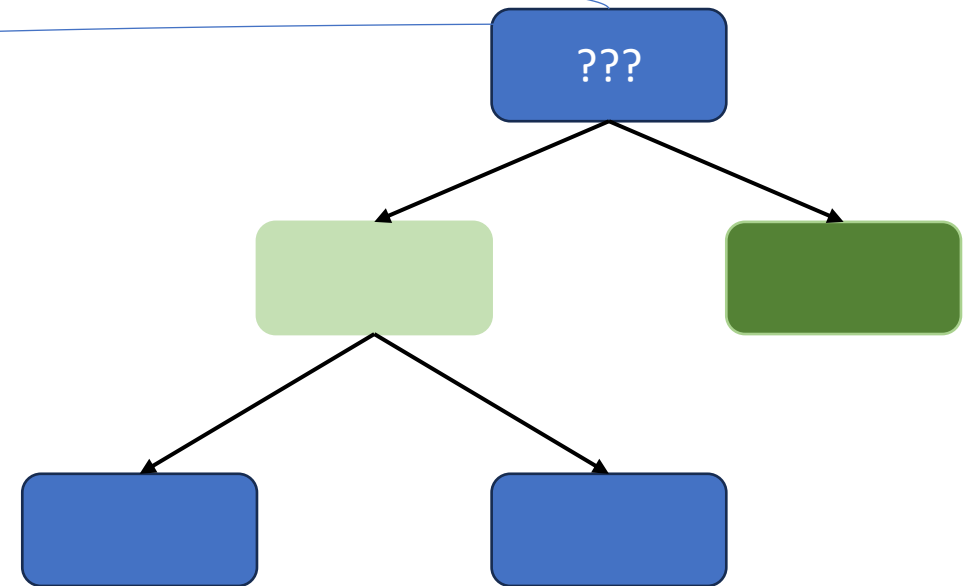
Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Building Tree with Gini Impurity

To make that decision, we'll start by looking at how well Like Math predicts whether or not someone Passes Exam.

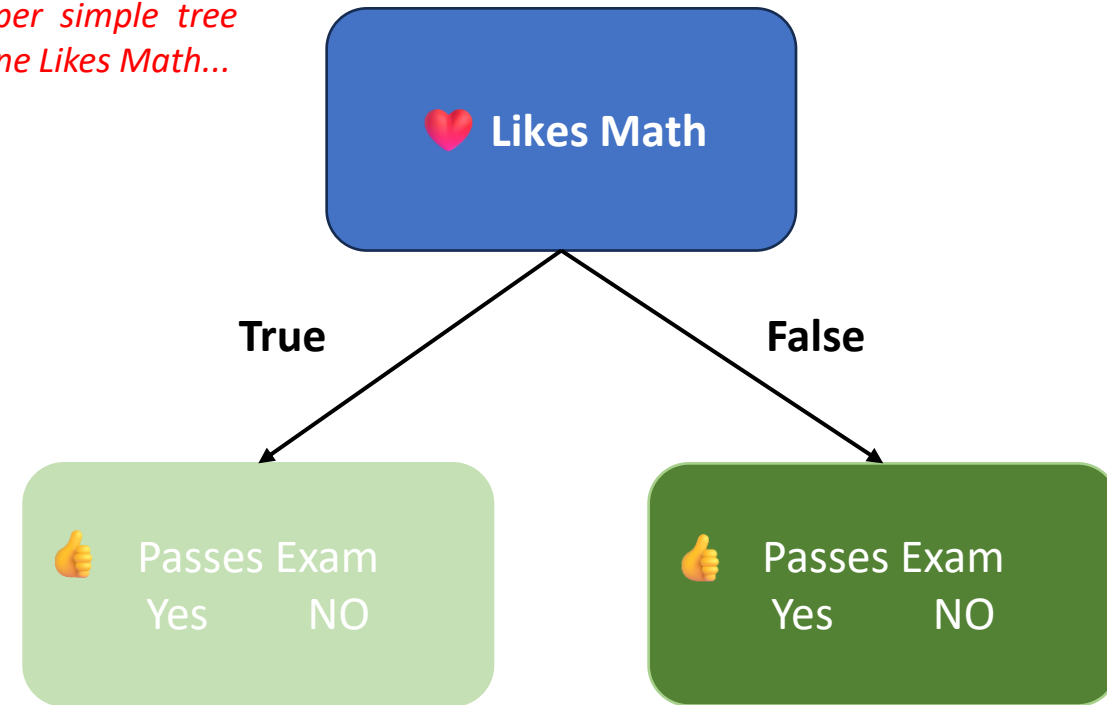
Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Building Tree with Gini Impurity

Now we'll make a super simple tree that only asks if someone Likes Math...

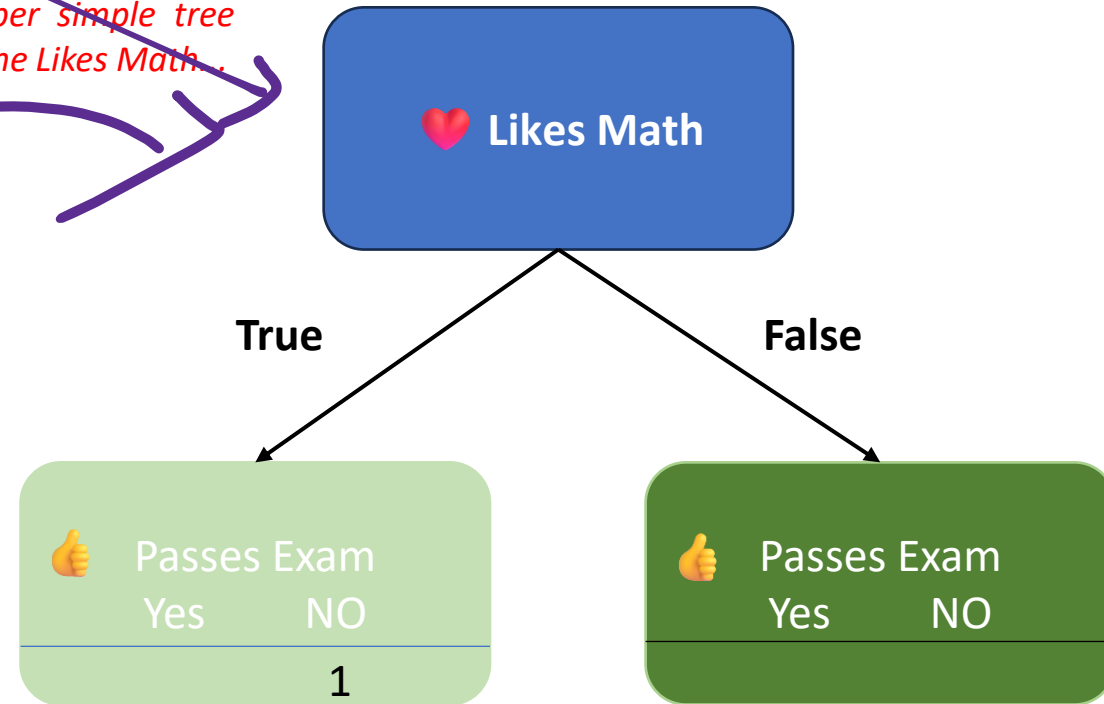
Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Building Tree with Gini Impurity

Now we'll make a super simple tree that only asks if someone Likes Math.

Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

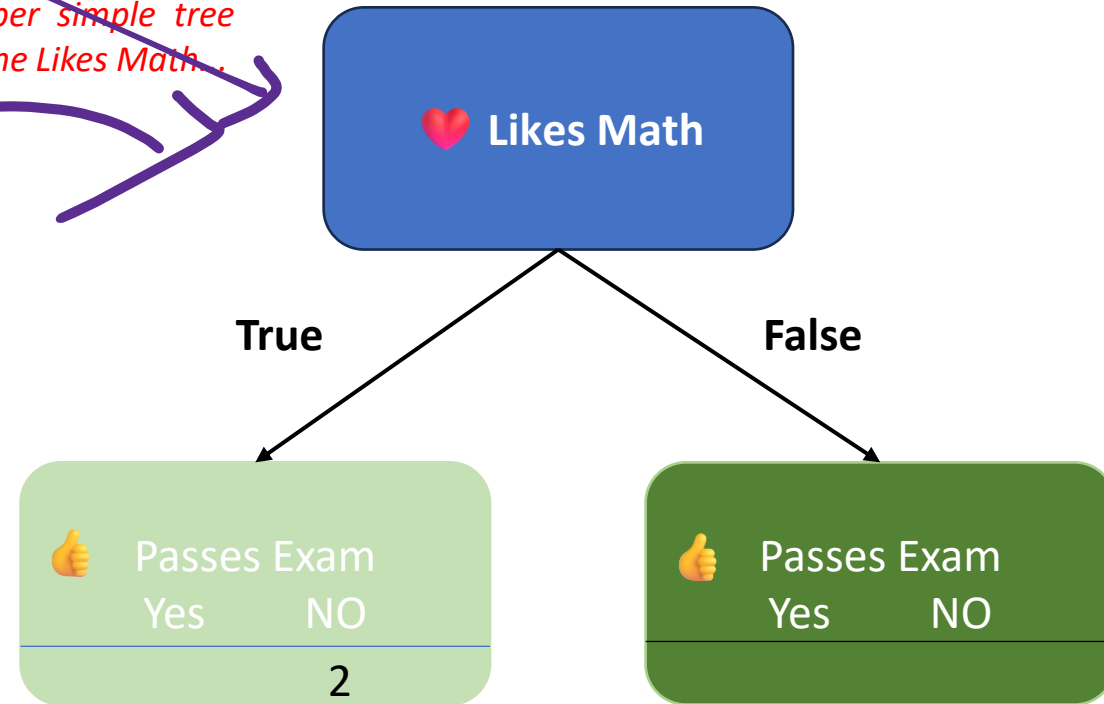


Run the data down the tree

Building Tree with Gini Impurity

Now we'll make a super simple tree that only asks if someone Likes Math.

Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

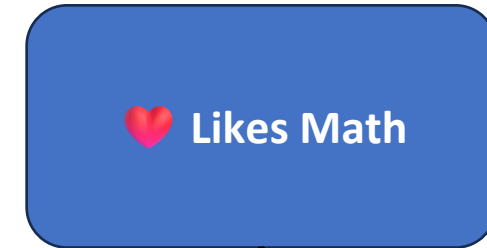


Run the data down the tree

Building Tree with Gini Impurity

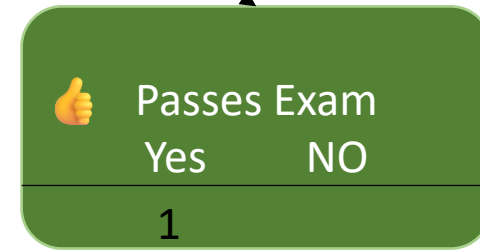
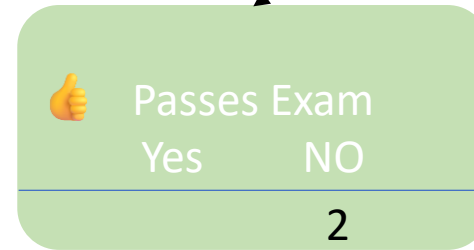
Now we'll make a super simple tree that only asks if someone Likes Math.

Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



True

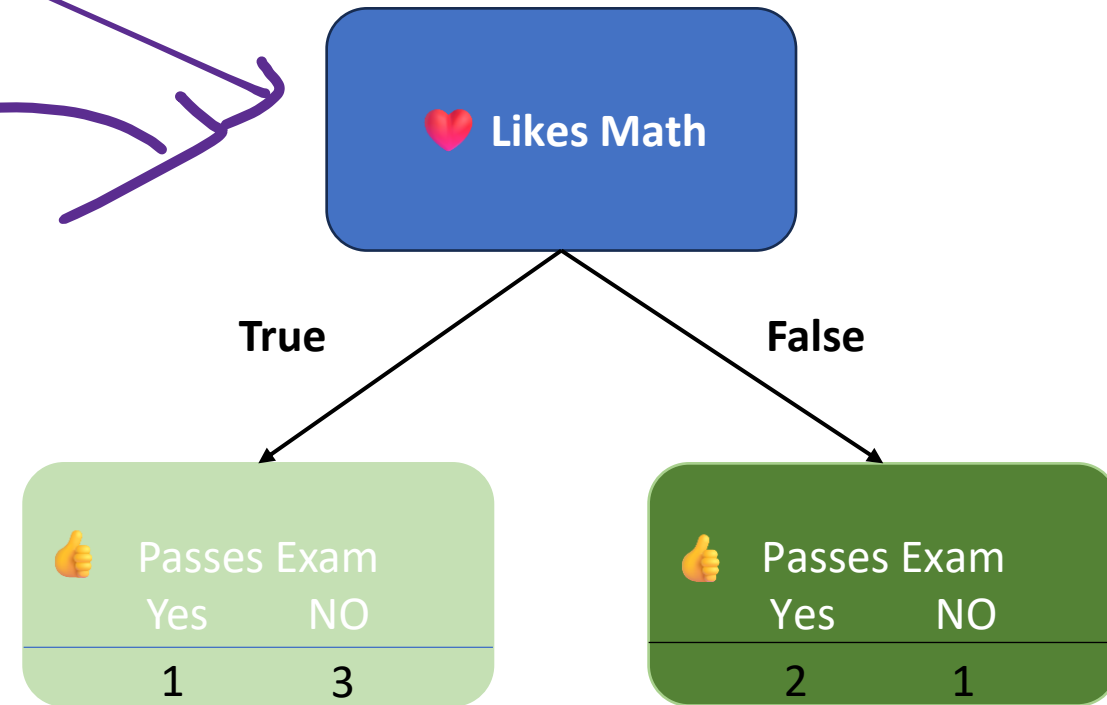
False



Run the data down the tree

Building Tree with Gini Impurity

Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

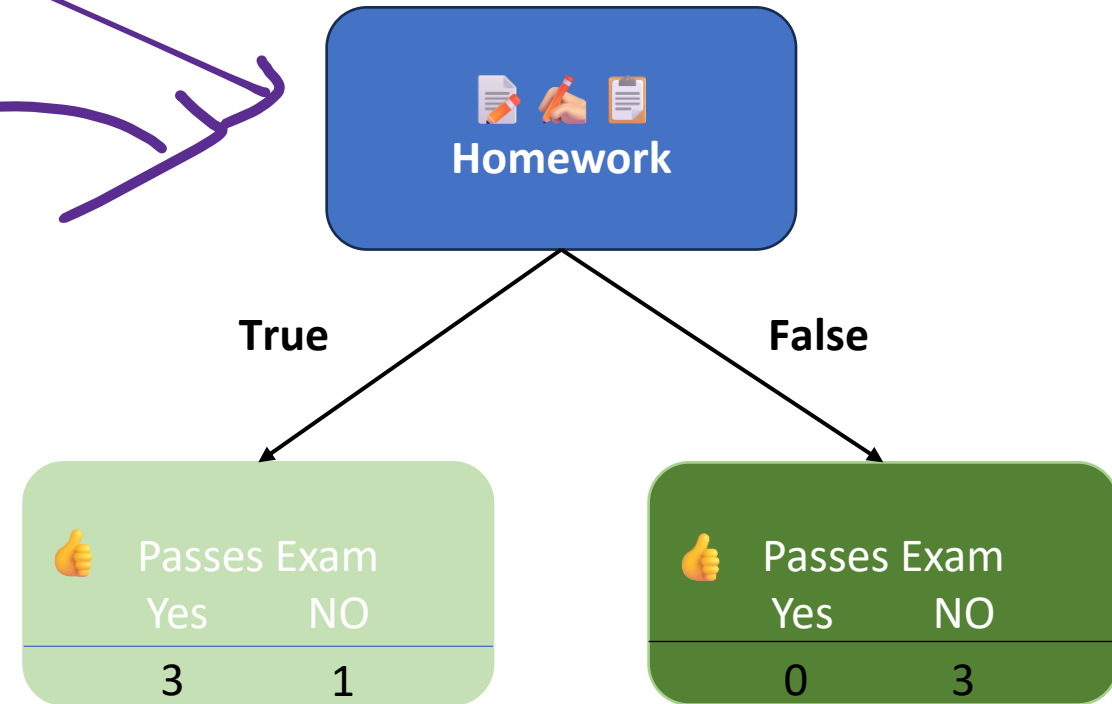


Likewise, we run the remaining rows down the tree, keeping track of whether or not each one Passes Exam

Building Tree with Gini Impurity

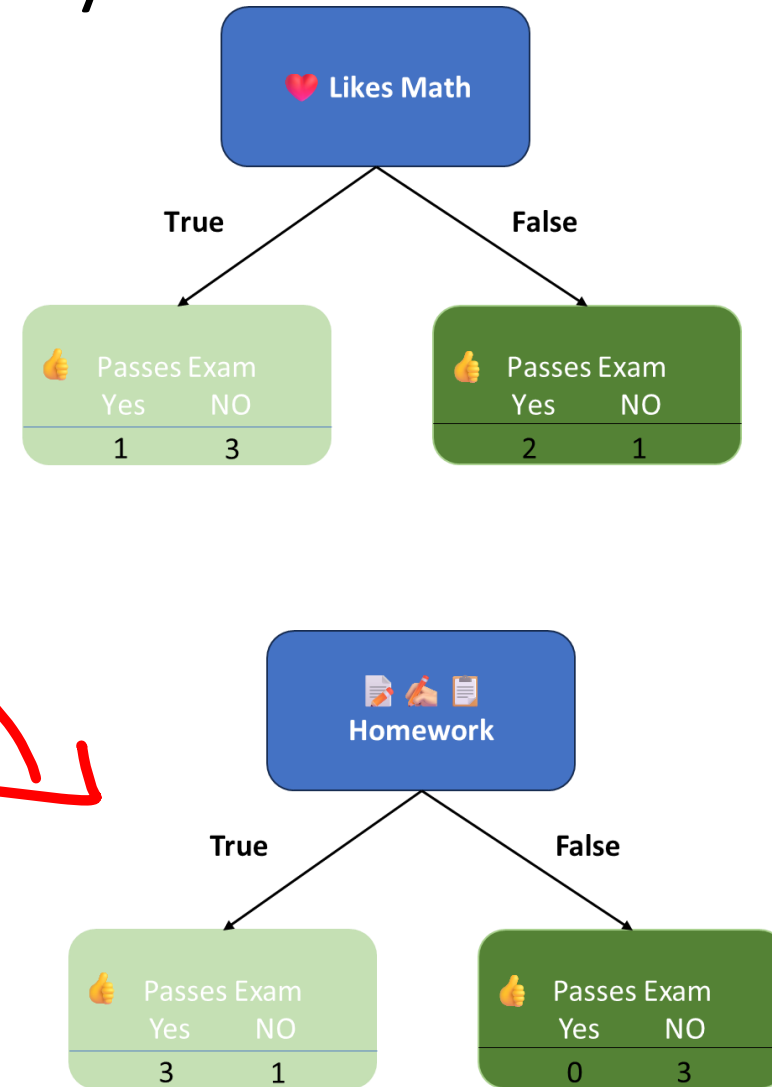
*We do the exact same thing with
Does Homework*

Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



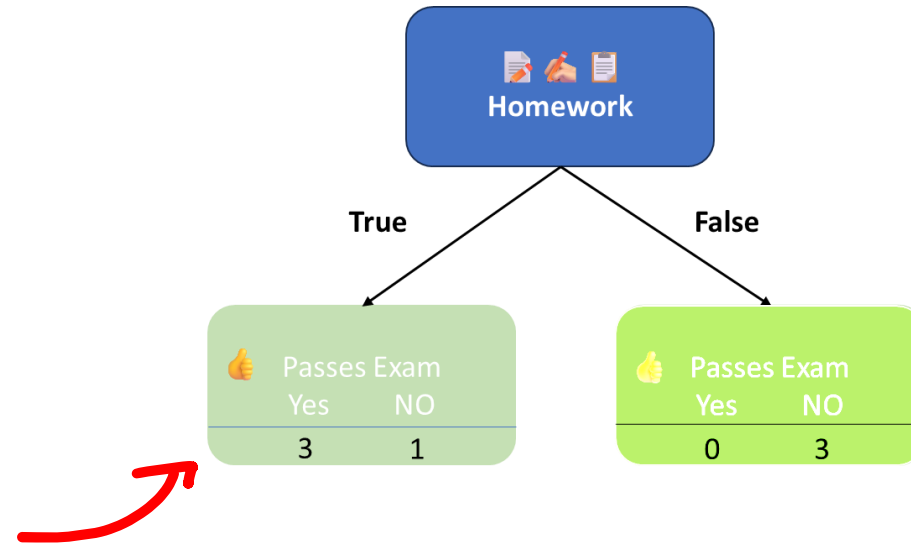
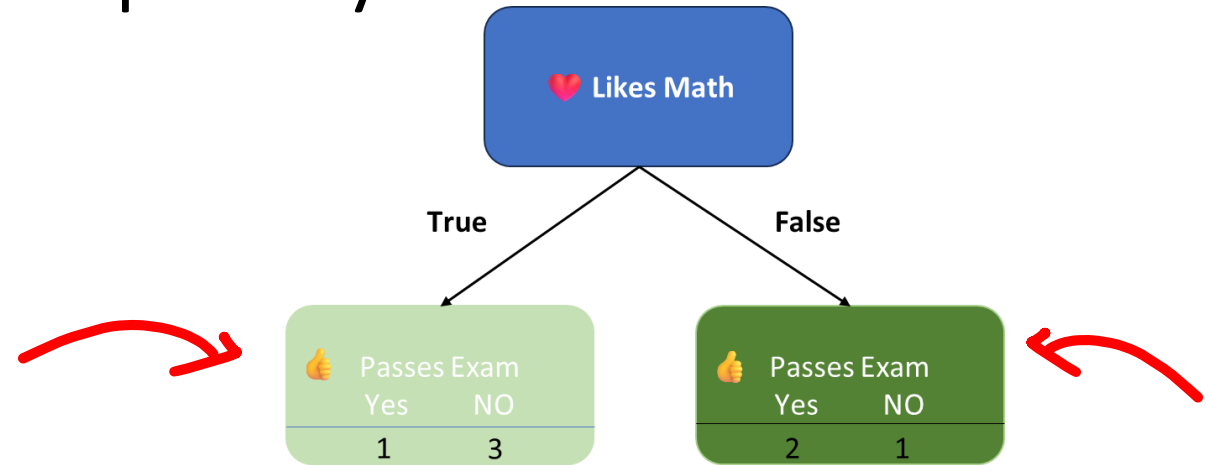
Building Tree with Gini Impurity

Looking at the two little trees, we see that neither one does a perfect job predicting who will and who will not Pass Exam.



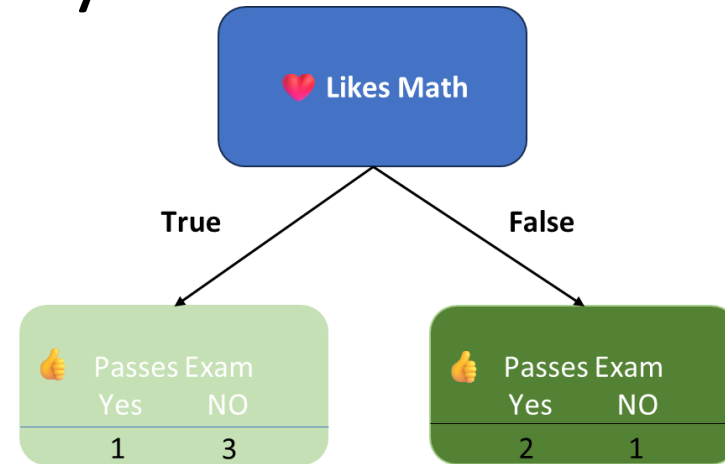
Building Tree with Gini Impurity

Specifically, these three Leaves contain mixtures of people that do and do not Pass Exam.

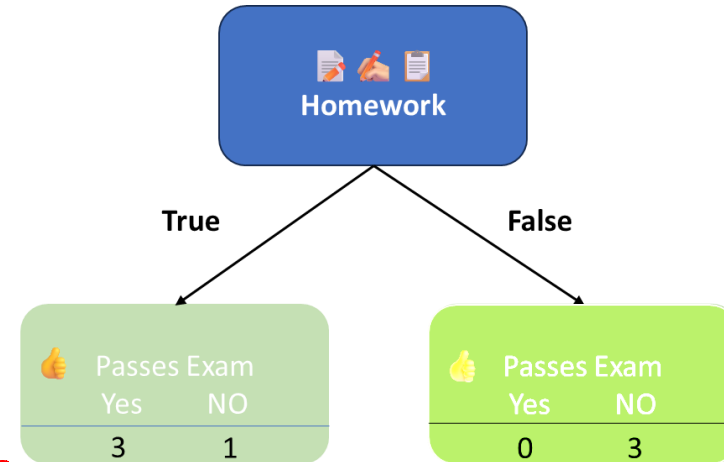


Building Tree with Gini Impurity

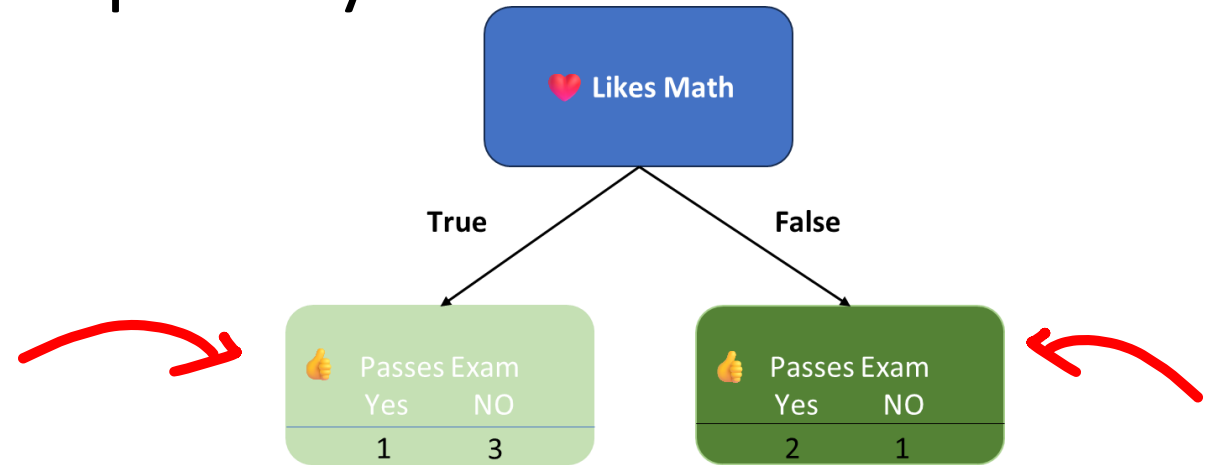
2 Impure Leaves



1 Impure Leaf

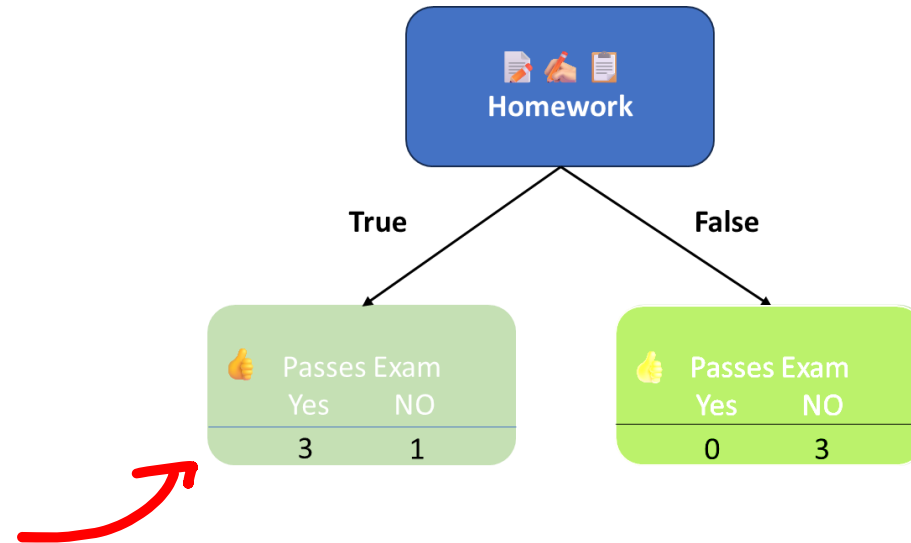


Building Tree with Gini Impurity



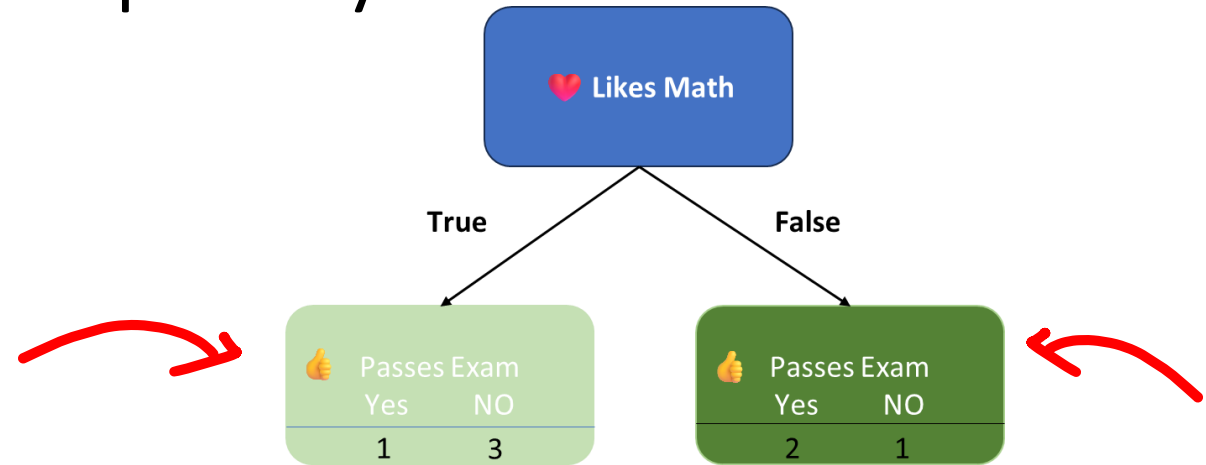
However, we could quantify the differences between Likes Math and Does Homework!

How?

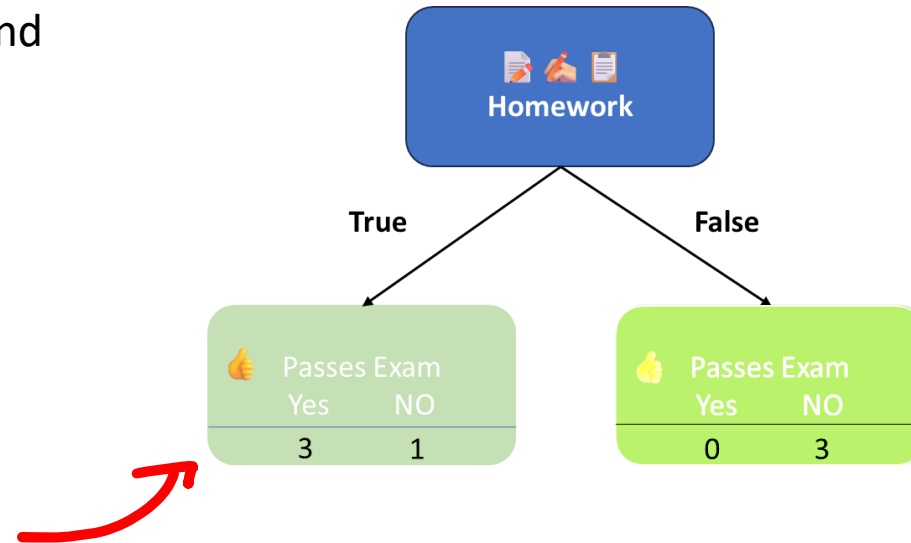


Building Tree with Gini Impurity

How?



One of the most popular methods is called **Gini Impurity**, but there are also fancy sounding methods like **Entropy** and **Information Gain**.



Building Tree with Gini Impurity

So! let's start by calculating the Gini Impurity for Likes Math

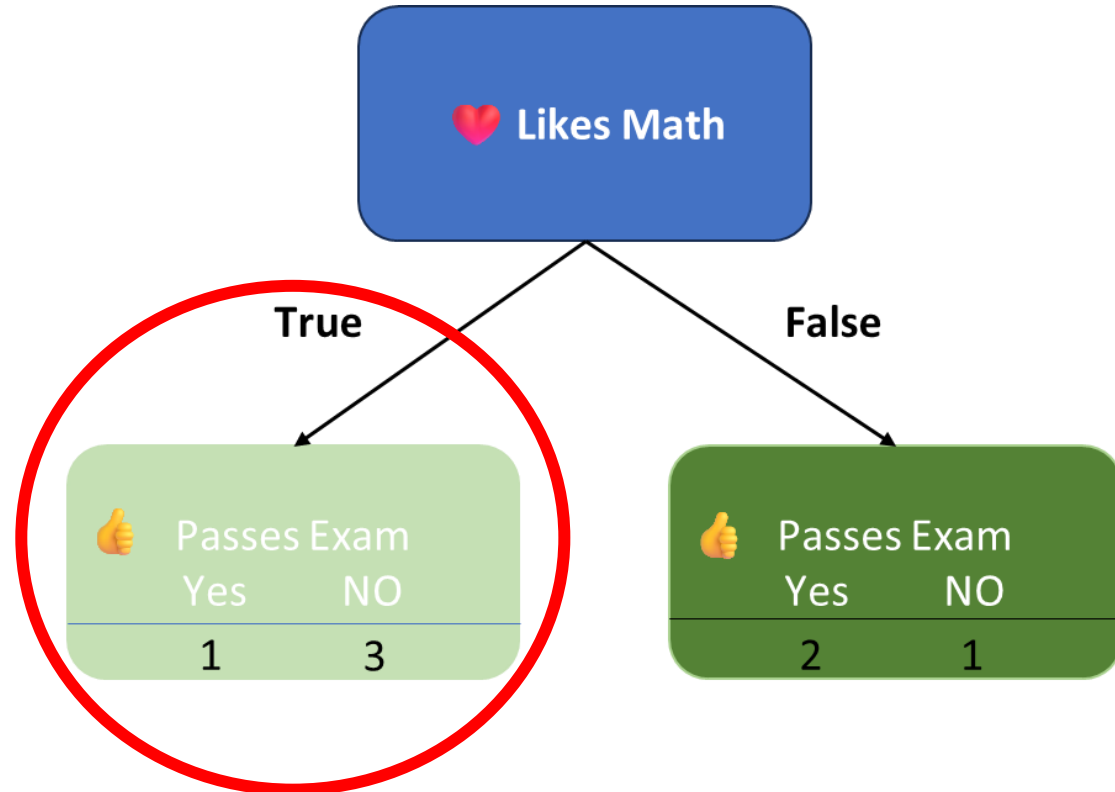
To calculate the Gini Impurity for Likes Math, we start by calculating the Gini Impurity for the individual Leaves.

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

Gini Impurity for a Leaf = 1 - (the probability of "Yes")² - (the probability of "No")²

$$= 1 - \left(\frac{1}{1+3}\right)^2 - \left(\frac{3}{1+3}\right)^2$$

And when we do the math, we get **0.375**.



Building Tree with Gini Impurity

So! let's start by calculating the Gini Impurity for Likes Math

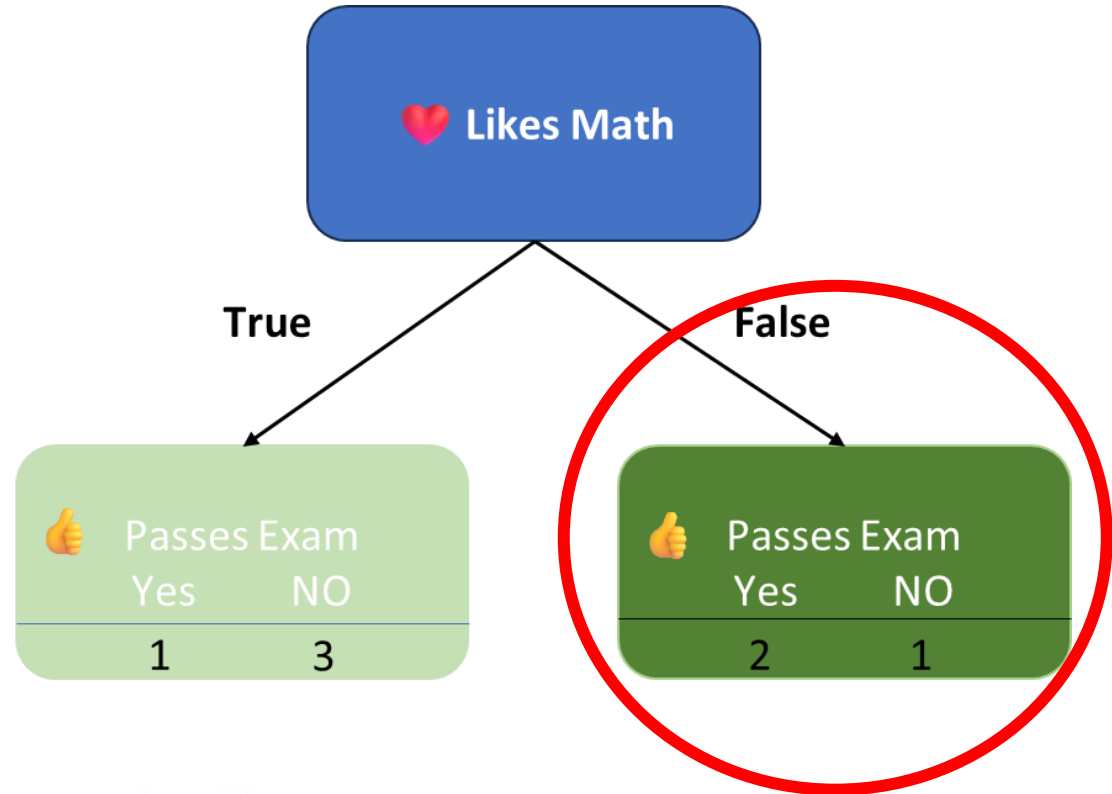
To calculate the Gini Impurity for Likes Math, we start by calculating the Gini Impurity for the individual Leaves.

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

Gini Impurity for a Leaf = $1 - (\text{the probability of "Yes"})^2 - (\text{the probability of "No"})^2$

$$= 1 - \left(\frac{2}{2+1}\right)^2 - \left(\frac{1}{2+1}\right)^2$$

And when we do the math we get **0.444**.

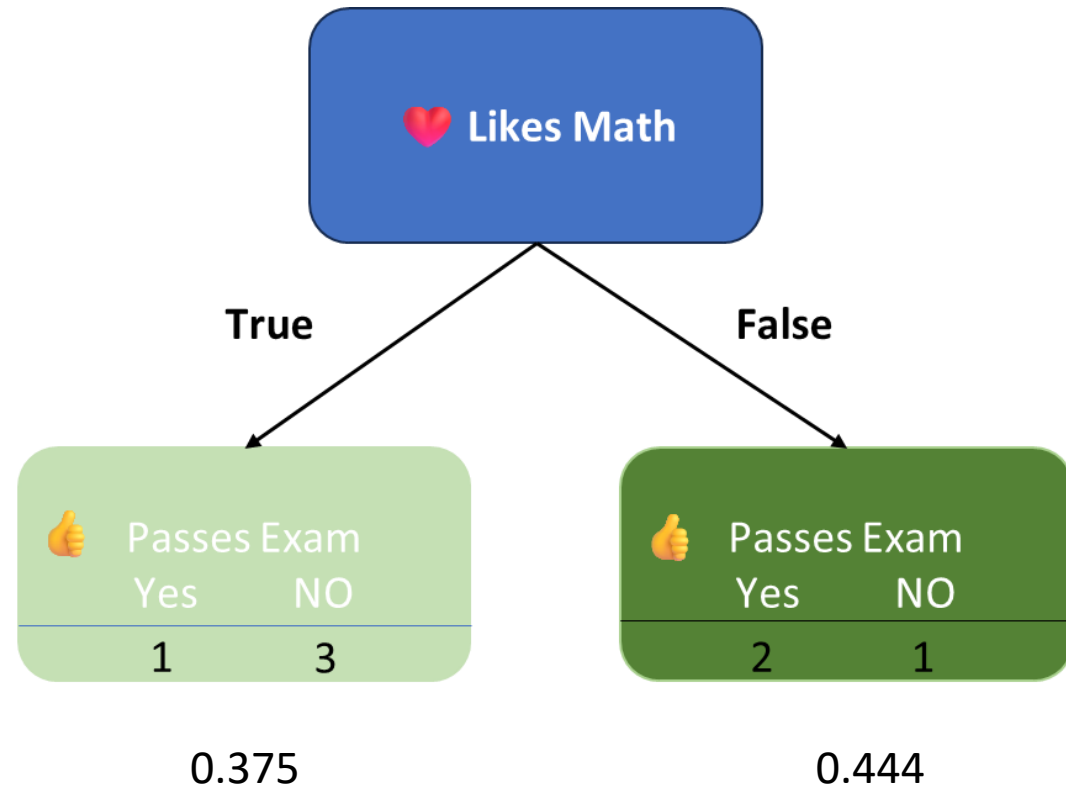


Building Tree with Gini Impurity

Now, because the Leaf on the left has 4 people in it...

...and the Leaf on the right only has 3 people in it...

...the Leaves do not represent the same number of people.

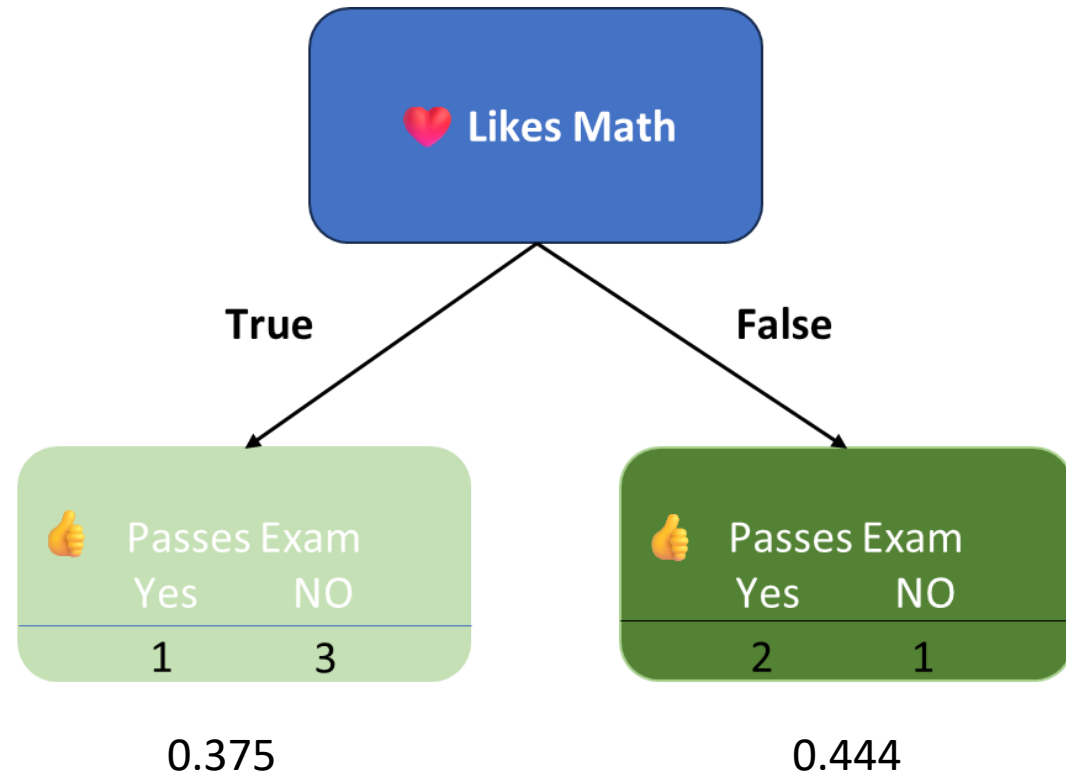


Building Tree with Gini Impurity

Now, because the Leaf on the left has 4 people in it...

...and the Leaf on the right only has 3 people in it...

...the Leaves do not represent the same number of people.



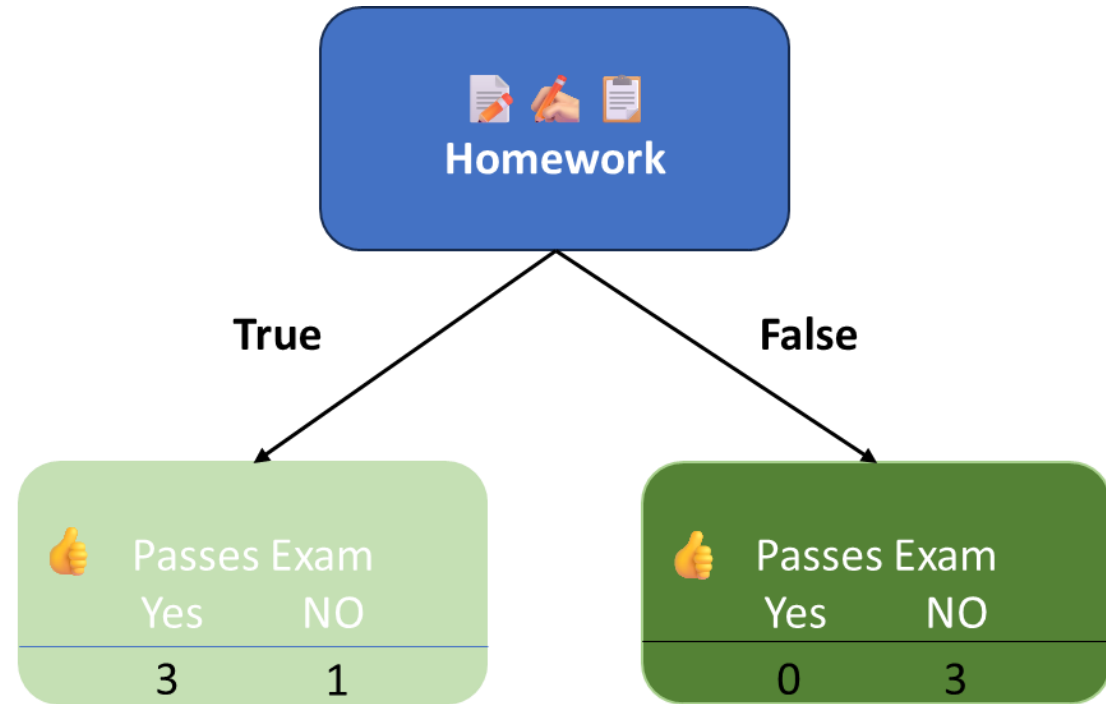
Total **Gini Impurity** = weighted average of **Gini Impurities** for the **Leaves**

$$= \left(\frac{4}{4+3} \right) 0.375 + \left(\frac{3}{4+3} \right) 0.444$$

The Gini Impurity for LIKES MATH = 0.405

Building Tree with Gini Impurity

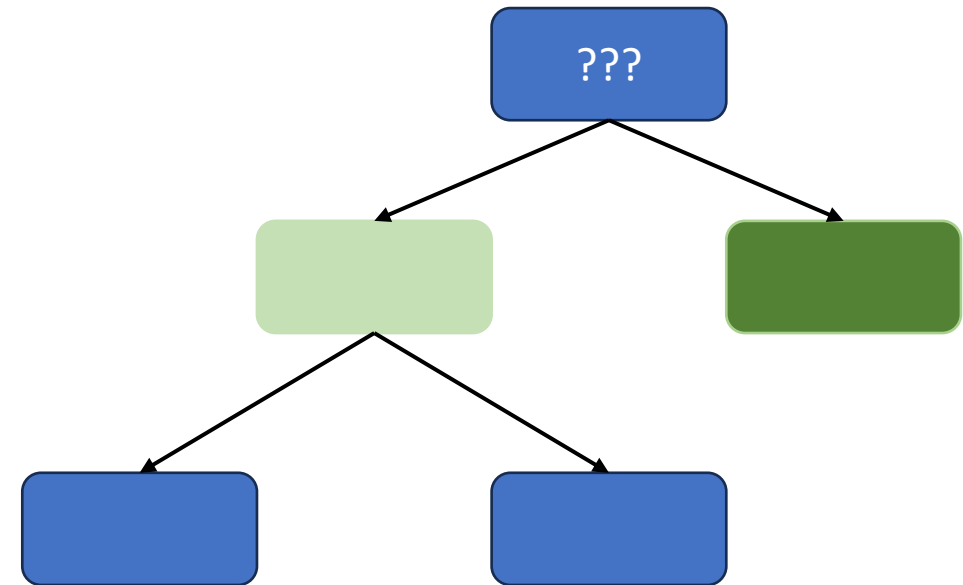
Likewise, the Gini Impurity for **Does Homework** is 0.214.



Building Tree with Gini Impurity

Now we need to calculate the Gini Impurity for *Study Hours*.

Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



However, because Study Hourse contains numeric data, and not just Yes/No values, calculating the Gini Impurity is a little more involved.

Building Tree with Gini Impurity

Now we need to calculate the Gini Impurity for *Study Hours*.

The first thing we do is sort the rows by **Study Hours**, from lowest value to highest value.

Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

Building Tree with Gini Impurity

Now we need to calculate the Gini Impurity for *Study Hours*.

Likes Math	Does Homework	Study Hours	Passes Exam
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

9.5	→	Gini Impurity = 0.429
15	→	Gini Impurity = 0.343
26.5	→	Gini Impurity = 0.476
36.5	→	Gini Impurity = 0.476
44	→	Gini Impurity = 0.343
66.5	→	Gini Impurity = 0.429